Assignment 1 Report

Gaussian Naive Bayes vs. Multinomial Naive Bayes

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1. Introduction

This project implements two types of Naive Bayes classifiers:

- Gaussian Naive Bayes for numerical data (Abalone dataset)
- Multinomial Naive Bayes for text classification (IMDB dataset)

The goal is to understand how Naive Bayes works for different data types and evaluate its performance.

2. Datasets

2.1 Abalone Dataset

- Used for predicting age groups of abalones.
- The original target (Rings) was converted into 3 classes:
  - Young (≤ 8)
  - Adult (9–11)
  - Old (> 11)
- The categorical feature Sex was removed.

2.2 IMDB Dataset

- Movie reviews labeled as:
  - Positive (pos)
  - Negative (neg)
- Text data was preprocessed before training.

3. Results

3.1 Gaussian Naive Bayes

- Accuracy: 67.82%
- The model shows moderate performance.
- It classifies the "Old" class well, but there is noticeable confusion between "Young" and "Adult", indicating overlapping feature distributions.

3.2 Multinomial Naive Bayes

- Accuracy: 85.62%
- The model achieves strong performance in sentiment classification.
- It effectively distinguishes between positive and negative reviews.

3.3 Comparison

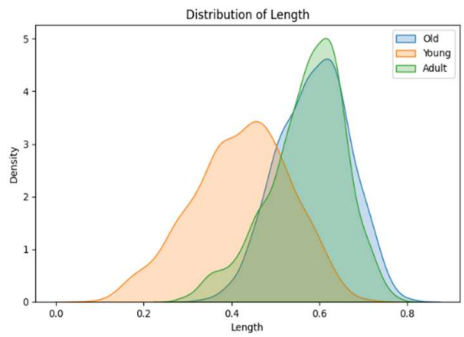
- Multinomial NB outperforms Gaussian NB.
- Gaussian NB is less effective due to the assumption of normal distribution, which does not fully fit the data.

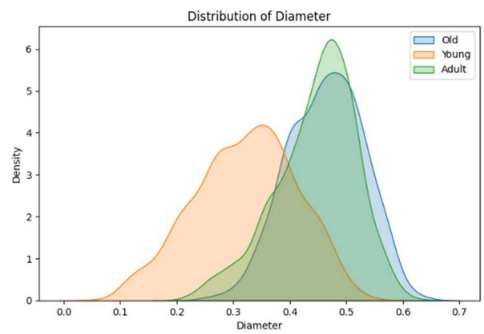
3.4 Conclusion

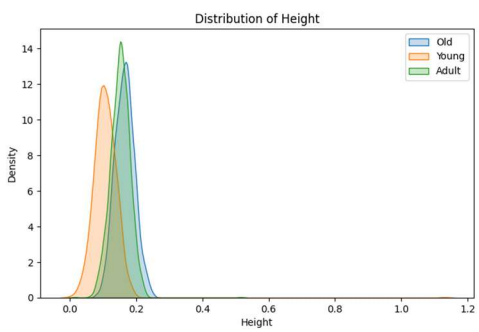
Naive Bayes performs well overall, especially for text data, while its performance on numerical data depends on how well the assumptions are satisfied.

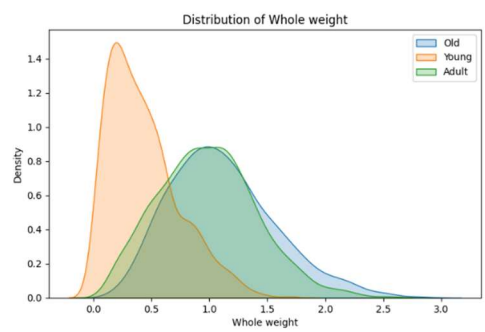
4. Visualization

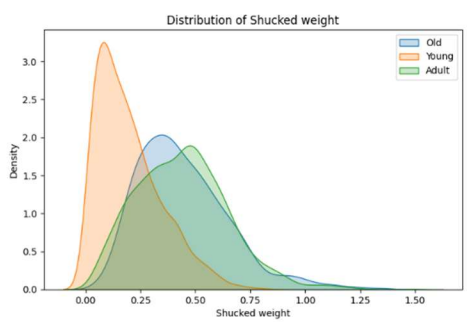
4.1 Feature Distributions

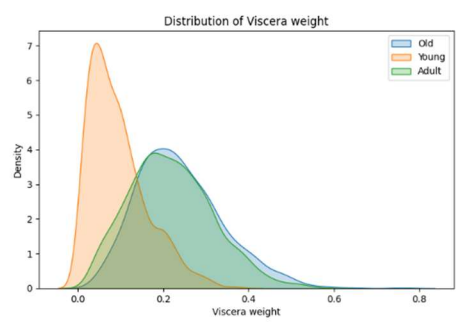


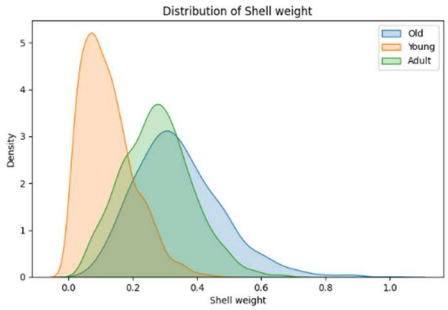
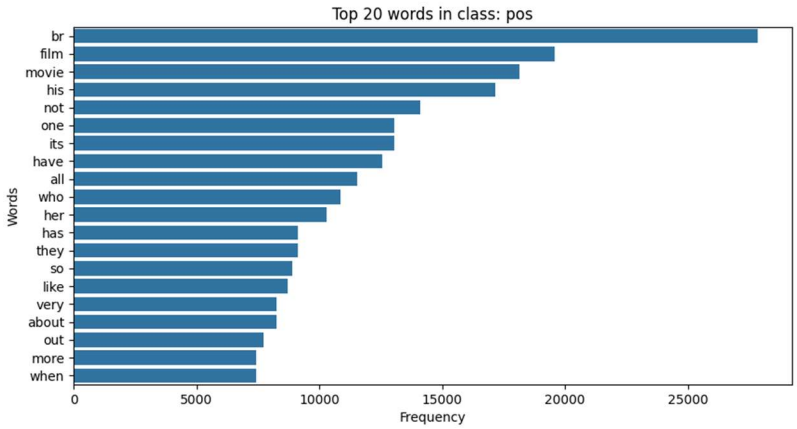
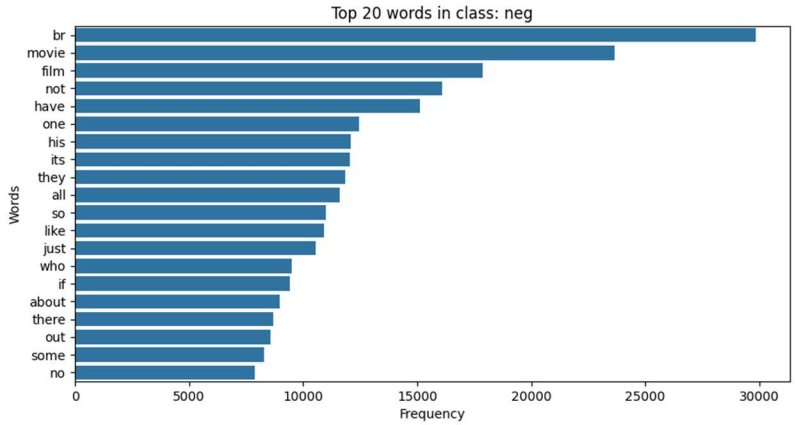












3.2 Top Words per Class